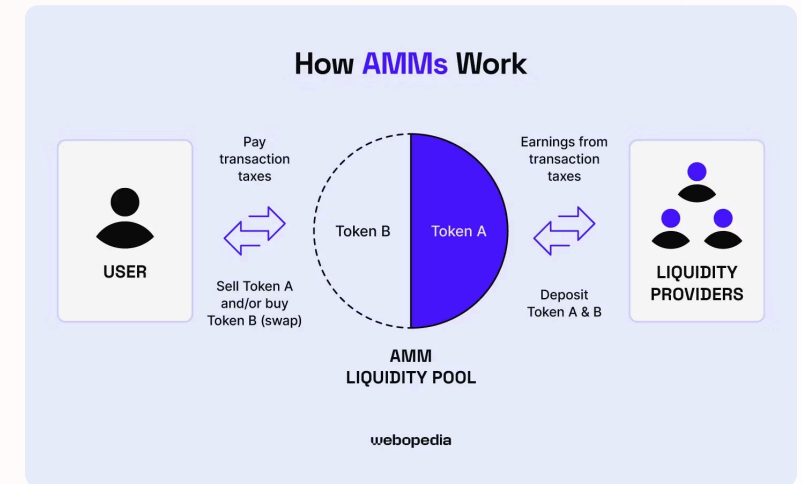


What is an Automated Market Maker?

An AMM is a smart contract that enables token swaps without traditional order books. Instead of matching buyers with sellers, AMMs use **liquidity pools** where users trade directly against pooled assets.

Think of it as two connected glasses filled with different liquids (tokens) of **equal USD value**. When someone pours more of one liquid in, the balance shifts, and the AMM automatically adjusts prices to restore equilibrium.

AMM - automated market maker



- ❏ Unlike traditional exchanges that need buyers and sellers to match orders, AMMs are **always ready to trade** as long as there's liquidity in the pool.

How AMMs Work: Three Perspectives

◆ Trader's Perspective

- Swaps tokens directly with the pool
- Pays a **small fee** (typically 0.3%)
- Example: Trader adds USDC to the pool and removes ETH, hence shifting the price

◆ Liquidity Provider's View

- Contributes equal USD value of both tokens
- Earns a portion of trading **fees** (usually 0.21%)
- Example: Provider adds \$1,000 of ETH + 1,000 USDC

◆ Platform Perspective

- Hosts the smart contracts (Uniswap, Orca, Cetus)
- Earn protocol fees or issue incentive tokens (usually 0.09%)
- Maintains the mathematical formulas governing swaps

This three-sided ecosystem creates a self-sustaining market where traders get liquidity, providers earn passive income, and platforms grow their ecosystems.

How to Provide ETH-USDC Liquidity on Uniswap | Earn Up to 50% APR - <https://youtu.be/busXMze0EpM>

Understanding Impermanent Loss

Impermanent loss occurs when token prices diverge from when you deposited them. It's the difference between:

1. The value if you just **held your tokens**
2. The value of your **liquidity position**

It's called "impermanent" because it can reverse if prices return to their original ratio. However, if you withdraw at the new price ratio, the loss becomes permanent.

						TOTAL
BEFORE ARB	10000	+	20	x	\$500	= \$20000
AFTER ARB	10488.09	+	19.07	x	\$550	= \$20976.59
					IMPERMANENT LOSS ←	\$23.41
	10000	+	20	x	\$550	= \$21000

Impermanent Loss: A Real Example

Initial Deposit

You provide **1 ETH (\$1,000) + 1,000 USDC** into a pool (total \$2,000)

Price Change

ETH price doubles from \$1,000 → \$2,000

HODLing Result

If you just held: 1 ETH (\$2,000) + 1,000 USDC = **\$3,000**

AMM Result

After rebalancing: ~0.71 ETH (\$1,420) + 1,420 USDC = **\$2,840**

When ETH's price moved from \$1,000 to \$2,000, you, as a liquidity provider, were still earning trading fees. In cases of sharp price swings, impermanent loss often outweighs the fees earned. But during a steady price move, the accumulated fees usually exceed the impermanent loss.

Also, with stable pools USDC-USDT, impermanent loss does not happen

❌ The **\$160 difference** is your impermanent loss. Even with trading fees, your position may be worth less than if you'd simply held your tokens.

Yield Opportunities & Risks

Potential Returns

Stablecoin Pairs

Low risk, low volatility, no impermanent loss

5-20% APY

Blue-chip Pairs (ETH-USDC, WBTC-USDC, SOL-USDC)

Moderate risk due to volatility, low impermanent loss

20-60% APY with incentives

Exotic Pairs (PEPE-USDC, and many more)

High risk, highly volatile, high impermanent loss

50-100%+ APY but unsustainable

Key Risks

- **Impermanent loss** - Pool positions may underperform simply holding tokens
- **Smart contract risk** - Code vulnerabilities could lead to fund loss
- **Liquidity migration** - Traders may move to new pools with better incentives,
- **Yield volatility** - APYs can drop as more LPs join a pool



Always research protocols thoroughly and consider starting with lower-risk pools as you learn the mechanics. Also time to time check APY and TVL* of the pools you are int

Personally, I recommend to use Blue chip pairs on biggest platforms (UNISWAP) to get great APR (50%) and have exposure with lower volatility to main assets of the crypto space

TVL* - total value locked